Project Title: Hospital Visit & Resource Management System

Project Overview

This project simulates a hospital database system that manages patient visits, doctor assignments, room allocation, service usage, billing, and payments. The goal is to create a structured and normalized relational database that ensures accuracy, automation, and efficient retrieval of hospital information.

Problem Statement

Hospitals handle interconnected data involving patients, staff, services, and financial transactions. Without a structured database, redundancy and inconsistencies occur. This project builds a normalized SQL system that maintains data integrity while supporting analytics, reporting, and automated workflows.

Project Objectives

Objective 1: Managing Hospital Data Efficiently
• Creating database tables and structure
• Adding and updating hospital records
• Retrieving patient and visit information
• Performing structured data operations

Objective 2: Connecting Hospital Information Across Departments
• Combining patient, doctor, and visit data
• Using different join techniques
• Generating integrated reports
• Cross-table analysis

Objective 3: Automating Hospital Workflows
• Creating stored procedures
• Automating billing logic
• Reducing manual operations

Objective 4: Handling Flexible and Modern Data Formats
• Working with JSON data
• Exploring NoSQL concepts
• Managing dynamic records

Objective 5: Ensuring Reliable and Secure Operations
• Maintaining database consistency
• Using transactions
• Creating triggers
• Designing views for reporting

Entity Relationship Diagram

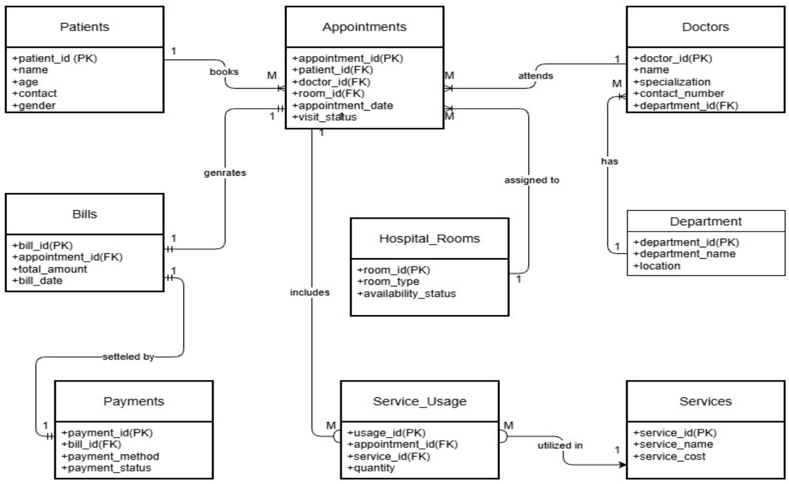

Figure 1: Entity Relationship Diagram for Hospital Visit & Resource Management System

Normalization Approach (Why 3NF?)

The database follows Third Normal Form (3NF) to eliminate redundancy and ensure consistency. Each table stores only relevant attributes, and non-key attributes depend only on the primary key. For example, departments are stored separately instead of repeating department names for every doctor. Services are stored once and reused across visits. This design prevents duplication, reduces storage waste, and avoids update anomalies. 3NF ensures accurate hospital records, easier maintenance, and scalable system growth.

Expected Outcome

The final outcome is a fully functional hospital database capable of tracking visits, services, billing, and payments while supporting analytics and automation. The system demonstrates real-world SQL skills including joins, aggregation, stored procedures, triggers, and transaction management.

Conclusion

This project provides a practical implementation of relational database design for hospital operations. It showcases structured problem solving, data integrity enforcement, and automation using SQL, preparing the system for real-world scalability and professional use.